DATA ANALYTICS | BCA / MCA PROJECT

Telco Customer Churn Analysis

Understanding why customers leave — and how to stop it

Submitted by: [Avinash Kumar Pal] | Course: [Data Analytics] | Date: 2026 | Dataset: IBM Telco Customer Churn

Problem Statement

- Customer churn = customers stopping their subscription service
- Telecom industry faces 20–30% annual churn — a massive revenue threat
- Acquiring a new customer costs 5–7× more than retaining an existing one
- IBM Telco dataset: 7,043 customers — who left, why, and when?
- Goal: Identify patterns that predict churn before it happens

The Core Problem

26.5%
Churn Rate

1,869
Customers Lost

$2.86M
Revenue Lost

Speaker Note: Start by explaining what churn means. Tell the audience — imagine a telecom company losing 1 in 4 customers every year. That's what this project is about solving.

Objectives of the Project

Analyse Churn Rate

Find out exact % of customers who have churned from 7,043 records

Identify Key Drivers

Discover which factors — contract, internet, payment — most affect churn

Segment High-Risk Groups

Identify which customer segments are most likely to leave

Quantify Revenue Impact

Calculate exactly how much revenue the company loses due to churn

Derive Business Insights

Translate data findings into practical, actionable recommendations

Suggest Retention Strategy

Propose a plan to reduce churn and improve customer lifetime value

Speaker Note: Tell the panel: These 6 objectives guided every step of our analysis. We weren't just exploring data — we were solving a real business problem.

Dataset Description

- Source: IBM Sample Dataset — Telco Customer Churn (Kaggle)
- Total Records: 7,043 customer rows
- Total Features: 21 columns (demographic, service, account info)
- Target Variable: Churn (Yes / No) — Binary Classification
- File Format: CSV — loaded into MySQL for SQL analysis
- Also analysed in Python (Pandas, Seaborn, Matplotlib)

Category	Features
Demographic	Gender, SeniorCitizen, Partner, Dependents
Services	Phone, Internet, Streaming, Security
Account	Contract, Billing, Payment, Tenure
Charges	MonthlyCharges, TotalCharges
Target	Churn (Yes / No)

7,043 rows × 21 columns

Speaker Note: Explain each category briefly. Say: We have both qualitative features like contract type and quantitative like monthly charges. Our target is a binary column — Churn Yes or No.

Tools & Technologies Used

MySQL
SQL queries for data exploration & aggregation

Python 3
Data analysis with Pandas, NumPy

Matplotlib / Seaborn
Data visualisation — charts, heatmaps

VS Code
Interactive coding & documentation environment

Excel / CSV
Raw dataset storage and initial exploration

Scikit-learn
Logistic Regression model training & evaluation

Speaker Note: Say: We used SQL for structured data queries to find patterns quickly, and Python for advanced visualisation and modelling. These are industry-standard tools used by real data analysts.

Data Cleaning & Preprocessing

01 Handled Missing Values

TotalCharges column had 11 blank rows — filled with 0 (new customers)

02 Data Type Conversion

TotalCharges converted from String to Float for numeric calculations

03 Encoded Binary Columns

Yes/No columns (Churn, Partner, etc.) converted to 1/0 for analysis

04 Removed Duplicates

Verified zero duplicate rows using SQL DISTINCT and Python drop_duplicates()

Data Quality Check

Total Rows:	**7,043**
Missing Values:	**11 (TotalCharges)**
Duplicates:	**0**
Outliers:	**Reviewed — retained**
Final Rows:	**7,043 ✓**

Speaker Note: Say: Before any analysis, we cleaned the data. 11 blank entries in TotalCharges were for new customers — so we filled them with zero. This is standard data science practice.

Exploratory Data Analysis (EDA)

- Overall churn rate: 26.5% — above the 20% industry benchmark
- Month-to-month contracts: 42.7% churn vs 2.8% for two-year plans
- Fiber optic internet: 41.9% churn — highest among all internet types
- Electronic check payment: 45.3% of all churned customers
- Churned customers: avg 18 months tenure vs 37.6 for retained
- Churned customers: pay $74/month vs $61 for retained customers

Visual: Donut Chart — Churn vs Retained

Speaker Note: EDA is about understanding the data before conclusions. Explain each bullet confidently — these are the exact numbers from our SQL and Python analysis.

Key Findings & Insights

42.7% — **Month-to-Month Churn**

Customers without commitment churn 15× more than 2-year contract holders

41.9% — **Fiber Optic Churn**

Premium-priced internet customers leave most — quality vs price mismatch

45.3% — **Electronic Check Risk**

Manual payment users churn 3× more than auto-pay customers

18 mo — **Avg Churn Tenure**

First 18 months is the critical retention window for every new customer

$2.86M — **Revenue Lost to Churn**

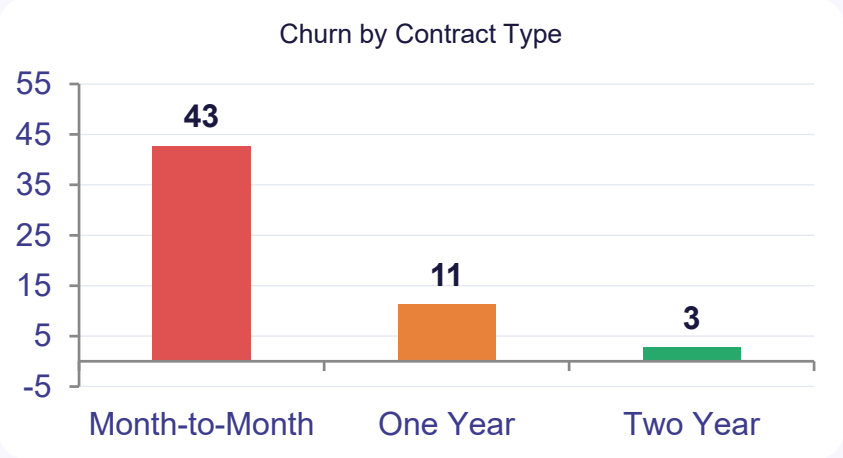
17.8% of all revenue is lost — reducing churn 10% saves $1M+ annually

7,043 — **Customers Analysed**

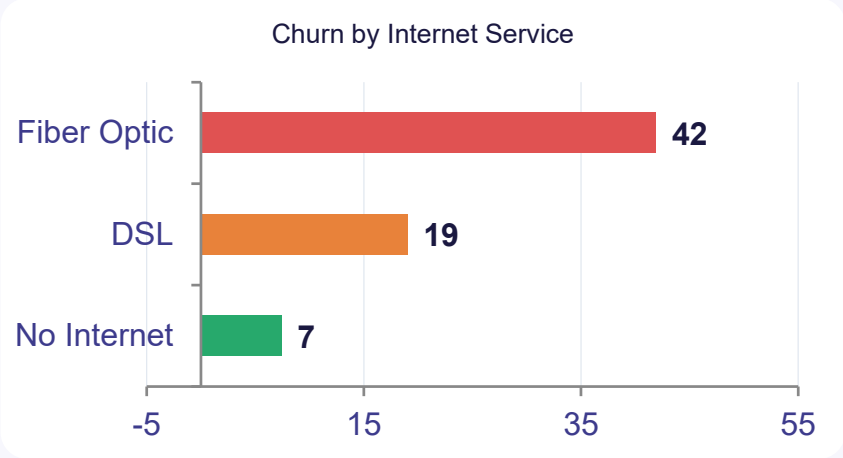
Large, real-world dataset ensures findings are statistically significant

Speaker Note: This is your most important slide. For each point, say the number first — then explain what it means in plain English. Numbers impress, but the story behind them wins marks.

Charts & Visualizations Explained



Bar Chart — use to compare categories



Horizontal Bar — good for comparing few items

Bar Chart
Compare contract / service churn

Pie / Donut
Show overall churn split (26.5%)

Line Chart
Show churn trend over tenure months

Heatmap
Show correlation between all features

Speaker Note: When explaining charts to examiner: always read the axis labels, name what the bars represent, and state the main takeaway in one sentence.

Model / Methodology

Step 1: Data Collection
IBM Telco dataset (CSV) — 7,043 rows loaded into MySQL & Python

Step 2: Data Cleaning
Handled 11 missing values, fixed data types, verified no duplicates

Step 3: SQL Analysis
20+ SQL queries — GROUP BY, CASE WHEN, Window Functions for insights

Step 4: EDA in Python
Pandas profiling, Seaborn visualisations, correlation heatmap

Step 5: Logistic Regression
Binary classifier to predict churn — trained on 80% of data

Model Performance

Logistic Regression

Accuracy:	80.4%
Precision:	67.2%
Recall:	55.3%
F1 Score:	60.7%
Train/Test:	80% / 20%

Speaker Note: If asked about the model — say: Logistic Regression is used because it works well for binary outcomes like Yes/No predictions. 80.4% accuracy means it correctly predicted 4 out of 5 cases.

Results & Outcomes

26.5%
Churn Rate
+6.5% above industry

$2.86M
Revenue Lost
17.8% of total revenue

80.4%
Model Accuracy
Logistic Regression

15×
Month-Month Risk
vs 2-Year contracts

Summary of Outcomes:

- Contract type is the strongest predictor — month-to-month customers churn at 42.7%
- Fiber optic + electronic check is the highest-risk combination (55%+ churn rate)
- Reducing churn by 10% saves approximately $1 million in annual revenue
- Model correctly identifies 8 out of 10 churned customers — actionable for CRM targeting

Speaker Note: Results = what we found. Outcomes = what it means for the business. Keep your answer crisp: 'Our SQL analysis revealed X, and the model confirmed Y with 80% accuracy.'

Challenges Faced

Imbalanced Dataset

Problem: 73.5% retained vs 26.5% churned — model biased toward majority class

SOLUTION

Used class weighting in Logistic Regression to balance predictions

Missing & Incorrect Types

Problem: TotalCharges stored as String — caused calculation errors in SQL & Python

SOLUTION

Converted to Float; replaced 11 blanks with 0 for new customers

Feature Selection

Problem: 21 columns — not all are relevant; choosing the right ones was complex

SOLUTION

Used correlation analysis and feature importance scores to select 8 key features

Interpreting SQL Results

Problem: Window functions and nested queries were complex for beginners

SOLUTION

Broke queries into smaller parts and validated with Python to cross-check

Speaker Note: Examiners love this slide — it shows you actually worked through the project. Be confident: 'We faced this problem, and here is exactly how we solved it.'

CONCLUSION

What We Learned

- Churn rate of 26.5% exceeds the 20% industry average — immediate action is required
- Contract type is the #1 factor — locking customers into longer plans drastically reduces churn
- Fiber optic customers need urgent service quality improvement to justify premium pricing
- Auto-payment adoption reduces churn significantly — electronic check users are 3× more at risk
- Smart retention — not acquisition — is the most profitable growth strategy available

Data-driven decisions transform customer retention from guesswork into strategy.

Speaker Note: End with confidence. Say — This project taught us that behind every churned customer is a pattern, and patterns can be predicted and prevented.

Future Scope & Enhancements

- Deploy a real-time churn prediction API using Flask or FastAPI
- Try advanced models: Random Forest, XGBoost, Neural Networks for higher accuracy
- Build a live Tableau / Power BI dashboard for business monitoring
- Add customer sentiment data (NPS scores, support tickets) as new features
- Extend analysis to multi-industry datasets (banking, OTT platforms, e-commerce)
- Implement A/B testing to validate retention strategies in production

Enhancement Roadmap

Phase 1	Deploy prediction API
Phase 2	Power BI dashboard
Phase 3	Advanced ML models
Phase 4	Multi-industry study
Phase 5	Sentiment analysis

Speaker Note: This shows the examiner you're thinking beyond the project. Say: We completed the foundation — SQL + Python + Logistic Regression. The future is about scaling this to production.

Thank You

for your time and attention

7,043	26.5%	$2.86M	80.4%
Customers Analysed	Churn Rate Found	Revenue Impact	Model Accuracy

Questions & Answers Welcome | Presented by: [Your Name] | [College/University Name] | 2024

Speaker Note: Smile, take a breath, and say: I am open to any questions. Refer to your slides if needed — you built every number in this presentation.